MA5211 Lie Theory

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These notes are based off **Dr. Emile Okada's** MA5211 Lie Theory materials.

This set of notes was last updated on March 5, 2026. If you would like to contribute a nice discussion to the notes or point out a typo, please send me an email at `thangpangern@u.nus.edu`.

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Chapter 1

From Lie Groups to Lie Algebras

1.1 Some Definitions

Definition 1.1 (Lie group). A Lie group is a group G which is also a finite-dimensional smooth manifold such that the group operations

$$G \times G \to G \text{ where } (x,y) \mapsto xy \quad \text{and} \quad G \to G \text{ where } x \mapsto x^{-1}$$

are smooth maps.

We give some remarks for Definition 1.1. Firstly, smooth means infinitely differentiable, i.e. $\in \mathcal{C}^\infty$. Also, an n-dimensional smooth manifold is locally $\mathbb{R}^n$ with $\mathcal{C}^\infty$ transition functions or change of local coordinates. In Definition 1.1, one may replace the smooth condition by a weaker one, say $\mathcal{C}^2$ or the stronger analytic condition. These turn out to be equivalent. This is related to Hilbert's fifth problem which asks whether locally Euclidean topological groups are Lie groups. The answer is affirmative.

Lastly, if the manifold is a complex manifold with group operations holomorphic, then we have a complex Lie group, which may be considered as a real Lie group of twice the dimension. In general when we talk about Lie groups, we mean real Lie groups.

Example 1.1 (real and complex vector space). Let V be a finite-dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$, viewed as an additive group $(V,+)$. It is known that V is a finite-dimensional smooth manifold. Well, if V is a real vector space of dimension n, choose a basis and identify $V \cong \mathbb{R}^n$. This gives V the structure of an n-dimensional smooth manifold. On the other hand, if V is a complex vector space of complex dimension n, choose a complex basis and identify $V \cong \mathbb{C}^n$. As a real manifold, $\mathbb{C}^n \cong \mathbb{R}^{2n}$ so V is a $(2n)$-dimensional smooth manifold over $\mathbb{R}$.

Here, the group operations are addition

$$m : V \times V \to V \quad \text{where} \quad m(x,y) = x + y$$

and inversion inversion

$$i : V \to V \quad \text{where} \quad i(x) = -x.$$

Example 1.2 (general linear group). Let V be a finite dimensional vector space over $\mathbb{R}$ or $\mathbb{C}$. Let $\mathrm{GL}(V)$ denote the set of invertible linear transformations of V. From MA2202 Algebra I, we have seen that $\mathrm{GL}(V)$ is a group under composition. Let $\mathrm{End}(V)$ denote the set of endomorphisms of V. This denotes the vector space of all linear transformations of V. As an open subset of $\mathrm{End}(V)$, $\mathrm{GL}(V)$ is a smooth manifold.

Since multiplication can be regarded as a quadratic polynomial map (using matrix representation through a standard basis) and taking inverse is a smooth operation by Cramer's rule in MA2001 Linear Algebra I, by Definition 1.1, $\mathrm{GL}(V)$ is a Lie group. In fact, it is given the following special name — the general linear group. If $\dim(V) = n$, then $\mathrm{GL}(V)$ may be identified with the group of $n \times n$ invertible matrices, and will be denoted by either $\mathrm{GL}(n, \mathbb{R})$ or $\mathrm{GL}(n, \mathbb{C})$.

Example 1.3 (Cartan's theorem). Cartan's theorem (Theorem 2.1) which will be formally discussed in due course states that a closed subgroup of a Lie group is also a Lie group.

Based on our discussion in Example 1.3, we have some other interesting examples. Lie groups which are closed subgroups of $\mathrm{GL}(V)$ are called matrix groups (Example 1.4).

Example 1.4 (matrix groups). We give some examples of matrix groups.

(i) The special linear group $\mathrm{SL}(n, \mathbb{R}) = \{A \in \mathrm{GL}(n, \mathbb{R}) : \det(A) = 1\}$

(ii) The orthogonal group $O(n) = \left\{A \in \mathrm{GL}(n, \mathbb{R}) : A^{\mathrm{T}} A = I_n\right\}$

(iii) The unitary group $U(n) = \{A \in \mathrm{GL}(n, \mathbb{C}) : A^* A = I_n\}$[1]

(iv) The special orthogonal group $\mathrm{SO}(n) = \mathrm{SL}\{n, \mathbb{R}\} \cap O(n)$

(v) The special unitary group $\mathrm{SU}(n) = \mathrm{SL}(n, \mathbb{C}) \cap U(n)$

Let $D = \mathbb{R}, \mathbb{C}, \mathbb{H}$ be one of the three division algebras over $\mathbb{R}$ (here, $\mathbb{H}$ denotes the quaternions), with the following involution:

$$\sigma = \begin{cases} \text{identity} & \text{if } D = \mathbb{R}; \\ \text{complex conjugation} & \text{if } D = \mathbb{C}; \\ \text{quaternionic conjugation} & \text{if } D = \mathbb{H}. \end{cases} \tag{1.1}$$

For notational convenience, we write $\sigma(\lambda) = \overline{\lambda}$ for $\lambda \in D$.

Definition 1.2 (sesquelinear form). As per our discussion above, let $D = \mathbb{R}, \mathbb{C}, \mathbb{H}$ be one of the three division algebras over $\mathbb{R}$ with the involution as in (1.1). Let $V = D^n = D \oplus \cdots \oplus D$ denote the direct sum of n copies of D. A map $V \times V \to D$ is a sesquelinear form $\langle \cdot, \cdot \rangle$ on V if it is additive in each variable, and for every

[1] Recall that A^* denotes the conjugate transpose of A.

$v_1, v_2 \in V$ and $\lambda \in D$,

$$\langle \lambda v_1, v_2 \rangle = \lambda \langle v_1, v_2 \rangle \quad \text{and} \quad \langle v_1, \lambda v_2 \rangle = \overline{\lambda} \langle v_1, v_@ \rangle .$$

Definition 1.3 (types of sesquelinear forms). As per our discussion in Definition 1.2, a sesquelinear form $\langle \cdot, \cdot \rangle$ on V is

(i) non-degenerate if $\langle v_1, v \rangle = 0$ for all $v \in V$, then $v_1 = 0$;

(ii) Hermitian if $\langle v_2, v_1 \rangle = \overline{\langle v_1, v_2 \rangle}$;

(iii) skew-Hermitian if $\langle v_2, v_1 \rangle = -\overline{\langle v_1, v_2 \rangle}$

Definition 1.4 (isometry group). As per our discussion in (1.2) and (1.3), given a non-degenerate Hermitian or skew-Hermitian form $\langle \cdot, \cdot \rangle$ on V, the isometry group of $\langle \cdot, \cdot \rangle$ is defined to be

$$G = \{ g \in \mathrm{GL}(V, D) : \langle g v_1, g v_2 \rangle = \langle v_1, v_2 \rangle \text{ for all } v_1, v_2 \in V \} .$$

Example 1.5 (non-degenerate Hermitian form). Let $D = \mathbb{R}, \mathbb{C}, \mathbb{H}$ be one of the three division algebras over $\mathbb{R}$ with the involution as in (1.1). Let V denote the direct sum of n copies of D. For non-negative integers p and q such that $p + q = n$, define the following non-degenerate Hermitian form on $V = D^n$ as follows:

$$\langle z, w \rangle = z_1 \overline{w}_1 + \cdots + z_p \overline{w}_p - z_{p+1} \overline{w}_{p+1} - \cdots - z_n \overline{w}_n ,$$

where $z = (z_1, \ldots, z_n)$ and $w = (w_1, \ldots, w_n)$. By Definition 1.4, its isometry group is denoted by $O(p,q), U(p,q), \mathrm{Sp}(p,q)$ for $D = \mathbb{R}, \mathbb{C}, \mathbb{H}$, and is respectively called the orthogonal, unitary, and symplectic groups of signature (p,q).

Example 1.6 (non-degenerate skew-symmetric bilinear form over $\mathbb{R}$). Let $D = \mathbb{R}$ be a division algebra over $\mathbb{R}$ with the involution as in (1.1). Let V denote the direct sum of n copies of D. A non-degenerate skew-symmetric bilinear form over $\mathbb{R}$ exists only on an even dimensional space. So, let $n = 2m$, and define the following form on $V = \mathbb{R}^{2m}$:

$$\langle x, y \rangle = x_1 y_{m+1} + \cdots + x_m y_{2m} - x_{m+1} y_1 - \cdots - x_{2m} y_m ,$$

where $x = (x_1, \ldots, x_{2m})$ and $y = (y_1, \ldots, y_{2m})$. Its isometry group is denoted by $\mathrm{Sp}(2m, \mathbb{R})$ and is called the real symplectic group of rank n. We may write the form

$$\langle x, y \rangle = x^{\mathrm{T}} J_m y \quad \text{where} \quad J_m = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} . \tag{1.2}$$

Then, its isometry group has the following matrix form:

$$\mathrm{Sp}(2m, \mathbb{R}) = \left\{ g \in \mathrm{GL}(2m, \mathbb{R}) : g^{\mathrm{T}} J_m g = J_m \right\} . \tag{1.3}$$

Example 1.7 (non-degenerate skew-Hermitian form). Let $D = \mathbb{H}$ be a division algebra over $\mathbb{R}$ with the involution as in (1.1). Let V denote the direct sum of n copies of D. Consider the following non-degenerate skew-Hermitian form on $V = \mathbb{H}^n$:

$$\langle z, w \rangle = z_1 j\overline{w}_1 + \cdots + z_n j\overline{w}_n,$$

where $z = (z_1, \ldots, z_n)$ and $w = (w_1, \ldots, w_n)$, and j is a pure quaternion, i.e. $\overline{j} = -j$. Its isometry group is denoted by $O^*(2n)$, known as the quaternionic orthogonal group.

Example 1.8 (symmetric and skew-symmetric forms). We have other variations, such as the isometry group of a non-degenerate symmetric form over $\mathbb{C}$, denoted by

$$O(n, \mathbb{C}) = \left\{ g \in \mathrm{GL}(n, \mathbb{C}) : g^{\mathrm{T}} g = I_m \right\},$$

and the isometry group of a non-degenerate skew-symmetric form over $\mathbb{C}$, denoted by

$$\mathrm{Sp}(2m, \mathbb{C}) = \left\{ g \in \mathrm{GL}(2m, \mathbb{C}) : g^{\mathrm{T}} J_m g = J_m \right\}, \tag{1.4}$$

where the matrix J_m was defined in (1.2).

So, we have the general linear group (Example 1.2), the isometry groups of symmetric or skew-symmetric (Examples 1.6 and 1.8) as well as Hermitian or skew-Hermitian forms (Examples 1.5 and 1.7) over $D = \mathbb{R}, \mathbb{C}, \mathbb{H}$. Those groups, together with their *sister groups* with the names of special linear groups, special orthogonal, and special unitary groups (Example 1.4), are all termed classical groups. We remark that an element of $\mathrm{Sp}(2m, \mathbb{R})$ (1.3) or $\mathrm{Sp}(2m, \mathbb{C})$ (1.4 automatically has determinant 1, so there is no such thing as a special symplectic group.

1.2 A Toolkit on Differential Geometry

Definition 1.5 (tangent space). Let M be a differentiable manifold. Then, a differentiable curve α is a differentiable map

$$\alpha : (a, b) \to M \quad \text{where } a, b \in \mathbb{R}.$$

We may assume that $0 \in (a, b)$. For $m \in M$, the tangent space of M at m, denoted by $T_m M$, is the vector space of velocity vectors $\alpha'(0)$, where α is a differentiable curve passing through $m \in M$.

An intrinsic definition of a tangent vector at m may be given as a derivation on the space of $\mathcal{C}^\infty$ functions on M around m, namely a linear functional D such that

$$D(f_1 f_2) = D(f_1) f_2(m) + f_1(m) D(f_2) \quad \text{where } f_i \in \mathcal{C}^\infty(M). \tag{1.5}$$

Define an operator

$$D_\alpha : \mathcal{C}^\infty(M) \to \mathbb{R} \quad \text{where} \quad D_\alpha(f) = \left. \frac{d}{dt} \right|_{t=0} (f \circ \alpha)(t).$$

Then, D_α is a derivation since it is $\mathbb{R}$-linear and it satisfies (1.5), and in fact D_α depends only on the velocity vector $\alpha'(0)$ by the chain rule. Now, let $f : M \to N$ be a smooth map and $m \in M$. For a tangent vector $X \in T_m M$, choose any smooth curve α with $\alpha(0) = m$ and $\alpha'(0) = X$. Define a linear map

$$(df)_m = T_m M \to T_{f(m)} N$$

by requiring that

$$(df)_m(X) = \left.\frac{d}{dt}\right|_{t=0} (f \circ \alpha)(t),$$

i.e. $(df)_m(X)$ is the velocity at $t = 0$ of the curve $f(\alpha(t))$ in N. This definition is independent of the choice of α with the same initial velocity X, and the resulting map $(df)_m$ is called the differential of f at m.

Definition 1.6 (differentiable vector field). Let M be a differentiable manifold. A differentiable vector field v on M is a differentiable section of the tangent bundle of M. That is, in a differentiable way,

$$M \ni m \mapsto v(m) \in T_m(m).$$

Definition 1.7 (integral curve). Let M be a differentiable manifold and v be a differentiable vector field on M. An integral curve of v is a curve

$$\alpha : (a,b) \to M \quad \text{such that} \quad \alpha(t) = v(\alpha(t)) \text{ for } t \in (a,b).$$

It is well-known that the first-order differential equation with initial condition $\alpha(t) = v(\alpha(t))$ where $\alpha(0) = m$ has a unique solution in a neighbourhood of 0, which we denote by $\Phi(t,m)$. Here, $t \in (a,b)$ for some $a,b \in \mathbb{R}$ and we may assume that (a,b) is the maximal interval on which the solution curve exists. Thus, we have a map

$$\Phi : (a,b) \times M \to M.$$

We have $\Phi(0,m) = m$ and

$$\Phi(t, \Phi(s,m)) = \Phi(t+s,m) \tag{1.6}$$

whenever either side is well-defined. This is because for a fixed s, each of them is the integral curve of v passing through $\Phi(s,m)$. If we write

$$\Phi_t : M \to M \quad \text{where} \quad m \mapsto \Phi(t,m),$$

then we have $\Phi_0 = \mathrm{id}$ and (1.6) may be rewritten as $\Phi_t \Phi_s = \Phi_{t+s}$, whenever either side is well-defined. The map Φ_t will be called the local flow associated with the vector field v.

Definition 1.8. Let G be a group and M be a set. We say that G acts on M or M is a

G-space if there exists a map

$$\Phi : G \times M \to M$$

such that $\Phi(e,x) = x$ and $\Phi(g,\Phi(h,x)) = \Phi(gh,x)$ for $x \in M$ and $g,h \in G$. Equivalently, let $\mathrm{Bi}(M)$ denote the group of bijective maps on M with composition as the group operation. Then, we have a group homomorphism $G \to \mathrm{Bi}(M)$, where $g \mapsto \Phi_g$.

Now, let M be a G-space. For any $x \in M$, the subset $\{\Phi_g(x) : g \in G\}$ of M is called the orbit of x, and the subgroup $\{g \in G : \Phi_g(x) = x\}$ of G is called the isotropy group or the stabiliser of x. Then, M is the disjoint union of all distinct orbits.

Now, let G be a Lie group. We shall see that the tangent spaces at points of G look the same everywhere. More precisely, for each $x \in G$, we define the left translation

$$L(x) : G \to G \quad \text{where} \quad g \mapsto xg.$$

Now, let G be a Lie group. We shall see that the tangent spaces at points of G look the same everywhere. More precisely, for any $x \in G$, we define the left translation

$$L(x) : G \to G \quad \text{where} \quad g \mapsto xg.$$

Each $L(x)$ is a diffeomorphism and we have $L(xy) = L(x)L(y)$, where $x,y \in G$. We shall therefore say that G acts on G by left translation. The differential of $L(x)$ yields a linear bijection between the tangent spaces. That is,

$$l(x) = dL(x) : T_gG \to T_{xg}G.$$

Definition 1.9 (left invariant vector field). A differential vector field v on G is called left invariant if

$$v(L(x)g) = l(x)v(g) \quad \text{where } x,g \in G.$$

Proposition 1.1. Let G be a Lie group and v be a left invariant vector field on G. Then, the flow associated to v is left invariant. That is,

$$\Phi(t,xg) = x\Phi(t,g) \quad \text{where } x,g \in G.$$

Consequently, $\Phi(t,g)$ is globally defined. Furthermore, we have $\Phi(t,g) = g\phi(t)$, where $\phi(t) = \Phi(t,e)$, and

$$\phi(t+s) = \phi(t)\phi(s) \quad \text{where } t,s \in \mathbb{R}.$$

Proof. By Definition 1.7, $\alpha(t) = \Phi(t,g)$ is the solution curve of $\alpha(t) = v(\alpha(t))$, where $\alpha(0) = g$. We have

$$\frac{d}{dt}(x\alpha(t)) = l(x)\alpha(t) = l(x)v(\alpha(t)) = v(x\alpha(t)).$$

Since $x\alpha(0) = xg$, then $\Phi(t, xg) = x\Phi(t, g)$, where $x, g \in G$. Now, suppose $\Phi(t, e)$ is defined for $|t| < \varepsilon$. Then, by Definition 1.9, $\Phi(s, e)\Phi(t, e) = \Phi(t, \Phi(s, e))$ is well-defined for $|t|, |s| < \varepsilon$, and so $\Phi(t, e)$ is at least well-defined for $t < 2\varepsilon$. The result follows. □

Definition 1.10 (one-parameter subgroup). A one-parameter subgroup of G is a smooth group homomorphism $\phi : \mathbb{R} \to G$.

Theorem 1.1. There are canonical bijections among the following four objects:

(i) $\{$one-parameter subgroups ϕ in $G\}$

(ii) $\{X \in T_eG\}$

(iii) $\{$left-invariant vector fields on $G\}$

(iv) $\{$left-invariant global flows Φ on $G\}$

1.3 The Exponential Map

We denote $\mathfrak{g} = T_eG$, which is called the Lie algebra of G. At this point, $\mathfrak{g}$ is just a real vector space.

Definition 1.11 (exponential map). The exponential map is defined to be

$$\exp : \mathfrak{g} \to G \quad \text{where} \quad X \mapsto \phi(1).$$

Here, ϕ is the one-parameter subgroup with initial velocity $X \in T_eG$.

Note that if ϕ is the one-parameter subgroup with initial velocity $X \in \mathfrak{g}$, then for any $\lambda\mathbb{R}$, the map $t \mapsto \phi(\lambda t)$ is the one-parameter subgroup with initial velocity λX, and so $\exp(\lambda X) = \phi(\lambda)$ for $\lambda \in \mathbb{R}$. We may therefore write the one-parameter subgroup associated to $X \in \mathfrak{g}$ as $\phi(t) = \exp(tX)$. The vector X is called the infinitesimal generator of the one-parameter subgroup ϕ.

Example 1.9. Recall from Example 1.2 that $\mathrm{GL}(V)$ denotes the set of invertible linear transformations of V. We mentioned that $\mathrm{GL}(V)$ is open in $\mathrm{End}(V)$. Now, for any $X \in \mathrm{End}(V)$, define

$$e^X = \sum_{k=0}^{\infty} \frac{X^k}{k!}.$$

This infinite series converges absolutely in the operator norm. Let $\phi(t) = e^{tX}$. Then,

$$\frac{d\phi}{dt} = Xe^{tX} = e^{tX}X = \widetilde{X}(\phi(t)),$$

where $\widetilde{X}$ is the left invariant vector field associated with $X \in \mathfrak{g}$. Thus,

$$\exp(tX) = e^{tX} \quad \text{where } t \in \mathbb{R}.$$

Proposition 1.2. The differential of the exp map at $0 \in \mathfrak{g}$ is the identity. Consequently, there exists a neighbourhood U of $0 \in \mathfrak{g}$ and a neighbourhood $e \in G$ of V such that $\exp : U \to V$ is a diffeomorphism.

Note that as $\mathfrak{g}$ is a vector space, it is connected, and so the image of the exponential map is a connected set in the Lie group G and it contains a neighbourhood of identity.

Definition 1.12 (Lie group homomorphism). Let G and G' be Lie groups. A Lie group homomorphism is a smooth group homomorphism $\Phi : G \to G'$.

Proposition 1.3 (naturality). Let $\Phi : G \to G'$ be a Lie group homomorphism, and let $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$ be its differential at the identity element of G. Then,

$$\Phi(\exp(X)) = \exp(\phi(X)) \quad \text{where } X \in \mathfrak{g}.$$

We now march towards the definition of a Lie bracket on $\mathfrak{g}$. For each $x \in G$, conjugation by x gives a Lie group homomorphism. That is,

$$c(x) : G \to G \quad \text{where} \quad g \mapsto xgx^{-1}.$$

Clearly, we have $c(x)c(y) = c(xy)$ for $x, y \in G$. We shall say G acts on G by conjugation. Taking the differential of $c(x)$ at e, we obtain a linear map denoted by $\mathrm{Ad}(x) : \mathfrak{g} \to \mathfrak{g}$. As such,

$$\mathrm{Ad}(x)\,\mathrm{Ad}(y) = \mathrm{Ad}(xy) \quad \text{for } x, y \in G.$$

We say that G acts on $\mathfrak{g}$ by the adjoint representation. By a representation of G, we mean a linear action of G by invertible linear transformations on a vector space.

For $X \in \mathfrak{g}$, $\mathrm{Ad}(\exp(tX))$ is an invertible linear transformation of $\mathfrak{g}$, and it is in fact a one-parameter subgroup of $\mathrm{GL}(\mathfrak{g})$. We denote its infinitesimal generator by $\mathrm{ad}(X)$. Thus, $\mathrm{ad}(X)$ is a linear transformation of $\mathfrak{g}$ such that

$$ad(X)Y = \frac{d}{dt}\left(\frac{d}{ds}\exp(tX)\exp(sY)\exp(-tX)\Big|_{s=0}\right)\Big|_{t=0} \quad \text{where } Y \in \mathfrak{g}.$$

Conceptually one may therefore regard $ad : \mathfrak{g} \to End(\mathfrak{g})$ as the differential of $\mathrm{Ad} : G \to GL(\mathfrak{g})$.

Definition 1.13 (Lie bracket). For $X, Y \in \mathfrak{g}$, define

$$[X, Y] = \mathrm{ad}(X)Y.$$

It is called the Lie bracket of X with Y.

It is easy to see that $[X, Y] = 0$ if and only if $\exp(tX)\exp(sY) = \exp(sY)\exp(tX)$ for $t, s \in \mathbb{R}$.

Example 1.10 (commutator). Let $G = GL(V)$ and $\mathfrak{g} = End(V)$. We have

$$\mathrm{Ad}(x)(Y) = xYx^{-1} \quad \text{where } x \in \mathrm{GL}(V) \text{ and } Y \in \mathrm{End}(V).$$

Therefore,

$$[X,Y] = \frac{d}{dt}\left(\exp(tX)Y\exp(-tX)\right)\Big|_{t=0} = XY - YX \quad \text{where } X,Y \in End(V).$$

This particular Lie bracket is called the commutator.

From Definition 1.13, it is clear that $[X,Y]$ is linear in both variables. But actually, the Lie bracket is skew-symmetric (Proposition 1.4).

Proposition 1.4 (skew-symmetry of Lie bracket). For any $X,Y \in \mathfrak{g}$, we have $[X,Y] = -[Y,X]$.

Proof. We consider the following differentiable function of two variables with values in G:

$$f(t,s) = \exp(tX)\exp(sY)\exp(-tX)\exp(-sY)$$

We shall compute $\frac{\partial^2 f}{\partial t\,\partial s}\Big|_{t=s=0}$ in two ways.

We first make a simple observation about the differential of the multiplication map μ. At (e,e), the differential is a linear map $d\mu : \mathfrak{g}\times\mathfrak{g} \to \mathfrak{g}$. Since μ is the identity map when one of the variables is fixed at e, we see that $d\mu$ must take the form $d\mu(X,Y) = X+Y$, where $X,Y \in \mathfrak{g}$.

We shall compute $\frac{\partial}{\partial t}\left(\frac{\partial f}{\partial s}\Big|_{s=0}\right)\Big|_{t=0}$. Write $f(t,s) = \alpha(t,s)\exp(-sY)$. For each fixed t, $\alpha(t,s)$ is a curve in G passing through e. We have

$$\frac{\partial \alpha}{\partial s}\Big|_{s=0} = \mathrm{Ad}(\exp(tX))Y \quad \text{and} \quad \frac{\partial \exp(-sY)}{\partial s}\Big|_{s=0} = -Y$$

and so by the above observation, we see that

$$\frac{\partial f}{\partial s}\Big|_{s=0} = Ad(\exp(tX))Y - Y.$$

Thus $\frac{\partial}{\partial t}\left(\frac{\partial f}{\partial s}\Big|_{s=0}\right)\Big|_{t=0} = [X,Y]$.

On the other hand, by writing $f(t,s) = \exp(tX)\beta(s,t)$, we compute to get

$$\frac{\partial}{\partial s}\left(\frac{\partial f}{\partial t}\Big|_{t=0}\right)\Big|_{s=0} = \frac{\partial}{\partial s}\left(X - Ad(\exp(sY))X\right) = -[Y,X].$$

The assertion follows. □

Proposition 1.5 (differential of a Lie group homomorphism is a Lie algebra homomorphism). Let $\Phi : G \to G'$ be a Lie group homomorphism, and let $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$ be its differential at the identity element of G. Then, ϕ preserves the Lie brackets of $\mathfrak{g}$ and $\mathfrak{g}'$. That is,

$$\phi([X,Y]) = [\phi(X), \phi(Y)] \quad \text{where } X, Y \in \mathfrak{g}.$$

Proof. For $x, y \in G$, we have

$$\Phi\left(xyx^{-1}\right) = \Phi(x)\Phi(y)\Phi(x)^{-1}.$$

Taking $y = \exp(tY)$ and computing $\frac{d}{dt}\big|_{t=0}$, we obtain

$$\phi(Ad(x)(Y)) = Ad(\Phi(x))\phi(Y).$$

Taking $x = \exp(tX)$ and computing $\frac{d}{dt}\big|_{t=0}$, we obtain

$$\phi([X,Y]) = [\phi(X), \phi(Y)].$$

□

Applying Proposition 1.5 to the adjoint representation $Ad : G \to GL(\mathfrak{g})$, we obtain

$$\mathrm{ad}[X,Y] = [\mathrm{ad}(X), \mathrm{ad}(Y)],$$

where the bracket on the right hand side is that of $End(\mathfrak{g})$ (the commutator). Namely,

$$\mathrm{ad}[X,Y] = \mathrm{ad}(X)\,\mathrm{ad}(Y) - \mathrm{ad}(Y)\,\mathrm{ad}(X).$$

We therefore have

$$\mathrm{ad}[X,Y](Z) = [[X,Y],Z] = [X,[Y,Z]] - [Y,[X,Z]],$$

for $Z \in \mathfrak{g}$. In view of the skew-symmetric property of the Lie bracket (Proposition 1.4), the above is equivalent to (and traditionally written as)

$$[[X,Y],Z] + [Y,[Z,X]] + [[Z,[X,Y]]] = 0 \quad \text{where } X, Y, Z \in \mathfrak{g}. \tag{1.7}$$

This is called the Jacobi identity.

Definition 1.14 (Lie algebra). A Lie algebra is a vector space $\mathfrak{g}$, together with a bilinear operation $[\cdot,\cdot]$ called bracket, which is skew-symmetric and which satisfies the Jacobi identity (1.7).

Definition 1.15 (Lie algebra homomorphism). Let $\mathfrak{g}$ and $\mathfrak{g}'$ be Lie algebras. A Lie algebra homomorphism is a linear map $\phi : \mathfrak{g} \to \mathfrak{g}'$ which preserves the bracket operations of $\mathfrak{g}$ and $\mathfrak{g}'$.

Theorem 1.2. Let G be a Lie group. Then $\mathfrak{g} = T_eG$ is a Lie algebra under the Lie bracket. Also, a Lie group homomorphism $\Phi : G \to G'$ induces a Lie algebra homomorphism $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$.

1.4 Connected Lie Groups

The relationship between Lie groups and Lie algebras is cleanest when one works with connected Lie groups. We begin with a simple property of such a group (Proposition 1.6).

Proposition 1.6. Let G be a connected Lie group. Then G is generated by any neighbourhood of the identity.

Proof. Let U be a neighbourhood of $e \in G$ and set $V = U \cap U^{-1}$. Then V is still a neighbourhood of e and satisfies $V^{-1} = V$. Consider

$$H = \bigcup_{k \geq 1} V^k,$$

the set of finite products of elements of V. It is immediate that H is a subgroup of G. We claim H is open. Indeed, if $g \in H$, then $g \in V^k$ for some k, hence $gV \subseteq V^{k+1} \subseteq H$, and gV is a neighbourhood of g. The complement $G \setminus H$ is a union of left cosets of H, hence also open. So H is both open and closed. Since G is connected and $e \in H$, we conclude $H = G$. □

Theorem 1.3. Let $\Phi : G \to G'$ be a Lie group homomorphism, and assume G is connected. Write $\mathfrak{g} = T_eG$, $\mathfrak{g}' = T_{e'}G'$ and $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$. Then Φ is uniquely determined by ϕ.

Proof. The exponential map $\exp : \mathfrak{g} \to G$ is a diffeomorphism from some neighbourhood U of $0 \in \mathfrak{g}$ onto a neighbourhood V of $e \in G$. By naturality (Proposition 1.3),

$$\Phi(\exp(X)) = \exp(\phi(X)),$$

so Φ is already determined by ϕ on V. Since G is connected, V generates G by the Proposition 1.6, and a group homomorphism is fixed by its values on a generating set. □

In general, we may consider the connected component of G at e, denoted by G_0.

Proposition 1.7. Let G_0 denote the connected component of G containing e. Then, the following hold:

(i) G_0 is a closed normal subgroup of G

(ii) The coset decomposition of G_0 in G coincides with the decomposition of G into connected components

Proof. We first prove **(i)**. A connected component is always closed as the closure of a connected set is connected. To see why G_0 is a subgroup, note that the map $(x,y) \mapsto xy^{-1}$ is continuous, and the image of $G_0 \times G_0$ under it is connected and contains e, hence lies in G_0. So $xy^{-1} \in G_0$ for all $x,y \in G_0$. Normality follows similarly: for fixed $g \in G$, the set $\{gzg^{-1} : z \in G_0\}$ is connected, contains e, and therefore lies in G_0.

For **(ii)**, each coset of G_0 is connected (continuous translate of G_0), and the cosets partition G, so they are exactly the connected components. □

A Lie group has a neighbourhood of e diffeomorphic to $\mathbb{R}^n$, hence contained in G_0. In particular, $T_eG = T_eG_0$. Next, the image of exp is connected and contains e, hence lies in G_0; since it contains a neighbourhood of e, the subgroup generated by $\exp(\mathfrak{g})$ is precisely G_0. So one-parameter subgroups in G and in G_0 are the same.

Locally, multiplication can be expressed purely in terms of brackets via the Baker-Campbell-Hausdorff series as follows:

$$\exp(X)\exp(Y) = \exp\left(X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}([[X,Y],Y] - [[X,Y],X]) + \cdots\right),$$

for X,Y sufficiently close to 0.

Theorem 1.4. If G is a connected Abelian Lie group, then

$$G \cong \mathbb{T}^k \times \mathbb{R}^{n-k},$$

where $n = \dim \mathfrak{g}$ and $\mathbb{T}$ is the circle group and $\mathbb{T}^k$ is the k-dimensional torus.

Proof. Since G is Abelian, for any $X,Y \in \mathfrak{g}$ the product $\exp(tX)\exp(tY)$ is a one-parameter subgroup whose initial velocity is $X+Y$. Hence

$$\exp(tX)\exp(tY) = \exp(t(X+Y)),$$

so $\exp : \mathfrak{g} \to G$ is a Lie group homomorphism (with $\mathfrak{g}$ viewed as an Abelian Lie group). For connected G, the image of exp generates G, but since the image is already a subgroup, it must equal G. Therefore by the first isomorphism theorem, $G \cong \mathfrak{g}/\ker(\exp)$.

Because exp is locally injective near 0, $\ker(\exp)$ is a discrete subgroup of the vector space $\mathfrak{g}$, hence is a free Abelian group of finite rank. That is,

$$\ker(\exp) \cong \mathbb{Z}X_1 \oplus \cdots \oplus \mathbb{Z}X_k.$$

Taking the quotient of $\mathbb{R}^n$ by a rank-k lattice gives $\mathbb{T}^k \times \mathbb{R}^{n-k}$. □

Corollary 1.1. A compact connected Abelian Lie group is a torus $\mathbb{T}^k$.

Next, consider a Lie group homomorphism $\Phi : G \to G'$ with differential $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$. Note that if $\ker(\Phi)$ is discrete (equivalently Φ is locally injective near e), then

ϕ is injective: if $\phi(X) = 0$, then $\Phi(\exp(tX)) = e'$ for all t, so $\exp(tX) \in \ker(\Phi)$; for small t, this forces $X = 0$. On the other hand, if ϕ is surjective, then $\Phi(G)$ contains a neighbourhood of e' in G', because $\Phi(\exp(X)) = \exp(\phi(X))$ and $\exp(\mathfrak{g}')$ contains a neighbourhood of e'.

Proposition 1.8 (local isomorphism criterion). Let $\Phi : G \to G'$ be a Lie group homomorphism such that $\ker(\Phi)$ is discrete. Assume G is connected and $\dim G = \dim G'$. Then

$$G/\ker(\Phi) \cong (G')_0,$$

where $(G')_0$ is the identity component of G'.

Proof. By discreteness of $\ker(\Phi)$, ϕ is injective. Equal dimensions imply ϕ is an isomorphism, hence surjective, so $\Phi(G)$ contains a neighbourhood of e'. The image is a connected subgroup (because G is connected), therefore it must be exactly $(G')_0$. The quotient by the kernel identifies $G/\ker(\Phi)$ with the image. □

Note that a Lie group covering of a connected Lie group G is a Lie group homomorphism $\pi : \widetilde{G} \to G$ with $\widetilde{G}$ connected and $(d\pi)_e$ an isomorphism of Lie algebras. A standard example is a 2-to-1 covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$ with kernel $\{\pm I_2\}$.

Finally, in the other direction (from Lie algebras to Lie groups), one has the following fundamental existence/uniqueness results (Theorem 1.5).

Theorem 1.5. The following hold:

(i) Every finite-dimensional Lie algebra $\mathfrak{g}$ is the Lie algebra of some simply connected Lie group $\widetilde{G}$

(ii) If G is connected and simply connected, and $\phi : \mathfrak{g} \to \mathfrak{g}'$ is a Lie algebra homomorphism, then there exists a unique Lie group homomorphism $\Phi : G \to G'$ satisfying $(d\Phi)_e = \phi$

We investigate **(ii)** of Theorem 1.5. Take $G = G' = \mathbb{T}$ which is connected but not simply connected. We have $\mathfrak{g} = \mathfrak{g}' = \mathbb{R}$ so a Lie algbera homomorphism $\phi : \mathbb{R} \to \mathbb{R}$ is of the form $\phi(t) = \lambda t$ for some $\lambda \in \mathbb{R}$. It is clear that ϕ comes from a Lie group homomorphism $\Phi : \mathbb{T} \to \mathbb{T}$ if and only if $\lambda \in \mathbb{Z}$. In general, in order to integrate a Lie algebra homomorphism to a Lie group homomorphism, one needs an integrality condition. Of course, if G is simply connected, no such condition is required.

Chapter 2

Subgroups, Quotients and Homogeneous Spaces

2.1 Lie Subgroups and Immersed Submanifolds

Definition 2.1 (embedded submanifold). Let M be a smooth manifold of dimension n. A subset $N \subseteq M$ is an m-dimensional embedded submanifold if for every $p \in N$ there exists a coordinate chart $(U; x_1, \ldots, x_n)$ around p such that

$$N \cap U = \{x \in U : \ x_{m+1} = \cdots = x_n = 0\}.$$

Definition 2.2 (immersed submanifold). An m-dimensional smooth manifold N is an immersed submanifold of an n-dimensional smooth manifold M if there exists an injective smooth map $f : N \to M$ such that for every $p \in N$, the differential $(df)_p : T_pN \to T_{f(p)}M$ is injective. In this situation, f is called an immersion.

Note that for an immersion $f : N \to M$, the image $f(N)$ need not be an embedded submanifold of M.

Definition 2.3 (Lie subgroup). Let G be a Lie group. A Lie group H is called a Lie subgroup of G if the following hold:

(i) H is a subgroup of G in the abstract group sense, and

(ii) the inclusion map $i : H \hookrightarrow G$ is an immersion

If $H \hookrightarrow G$ is a Lie subgroup, then the induced map on tangent spaces at the identity gives an inclusion

$$\mathfrak{h} = T_eH \hookrightarrow T_eG = \mathfrak{g}.$$

The naturality of the differential (Proposition 1.3) means that for $Y \in \mathfrak{h}$, the image of Y under the exponential map from $\mathfrak{h}$ to H and the exponential map from $\mathfrak{g}$ to G are the

same. Furthermore, as $\mathfrak{h} \hookrightarrow \mathfrak{g}$ is a Lie algebra homomorphism, we see that for $Y_1, Y_2 \in \mathfrak{h}$, the bracket $[Y_1, Y_2]$ gives the same result whether one considers them as elements of $\mathfrak{h}$ or $\mathfrak{g}$. Thus, we may use the same notations exp and $[\cdot,\cdot]$ without confusion.

With this convention in mind, we may describe the Lie algebra $\mathfrak{h}$ as

$$\mathfrak{h} = \{Y \in \mathfrak{g} : \exp(tY) \subseteq H \text{ for all } t \in \mathbb{R}\}.$$

Furthermore, $\mathfrak{h}$ is a Lie subalgebra of $\mathfrak{g}$ in the sense that $[Y_1, Y_2] \in \mathfrak{h}$ for $Y_1, Y_2 \in \mathfrak{h}$.

Example 2.1. We have the orthogonal group

$$O(n) = \left\{g \in \mathrm{GL}(n, \mathbb{R}) : g^{\mathrm{T}} g = I_n\right\}.$$

Denote its Lie algebra by $\mathfrak{o}(n)$. We have

$$X \in \mathfrak{o}(n) \quad \text{if and only if} \quad \left((\exp(\lambda X))^{\mathrm{T}} \exp(\lambda X)\right) \in I_n \text{ for all } \lambda \in \mathbb{R}.$$

Taking the derivative with respect to λ, and then evaluating at $\lambda = 0$, we have $X^{\mathrm{T}} + X = 0$. So, X is skew-symmetric. Conversely, if $X^{\mathrm{T}} + X = 0$, then $\left((\exp(\lambda X))^{\mathrm{T}} \exp(\lambda X)\right) \in I_n$. Hence,

$$\mathfrak{o}(n) = \left\{X \in \mathcal{M}_n(\mathbb{R}) : X^{\mathrm{T}} = -X\right\}$$

which has dimension $\frac{n(n-1)}{2}$.

Example 2.2. We consider the real symplectic group

$$\mathrm{Sp}(2n, \mathbb{R}) = \left\{g \in \mathrm{GL}(2n, \mathbb{R}) : g^{\mathrm{T}} J_n g = J_n\right\}$$

with Lie algebra $\mathfrak{sp}(2n, \mathbb{R})$. Then, one can prove that

$$\mathfrak{sp}(2n, \mathbb{R}) = \left\{X \in \mathcal{M}_{2n}(\mathbb{R}) : X^{\mathrm{T}} J_n + J_n X = 0\right\}.$$

We may write it more explicitly as

$$\mathfrak{sp}(2n, \mathbb{R}) = \left\{X = \begin{pmatrix} A & B \\ C & -A^{\mathrm{T}} \end{pmatrix} \in \mathcal{M}_{2n}(\mathbb{R}) : B^{\mathrm{T}} = B \text{ and } C^{\mathrm{T}} = C\right\}$$

which is of dimension $2n^2 + n$.

Theorem 2.1 (Cartan's theorem). A closed subgroup H of a Lie group G is a Lie group and therefore a Lie subgroup of G.

It may be sown that a Lie subgroup H of a Lie group G is closed in G if and only if it is a submanifold. We now give an example of an immersed subgroup that is not closed (Example 2.3).

Example 2.3 (dense one-parameter subgroup of $\mathbb{T}^2$). Define $f : \mathbb{R} \to \mathbb{T}^2$ by

$$f(t) = \left(e^{2\pi i t}, e^{2\pi i \sqrt{2} t}\right) \quad \text{where } t \in \mathbb{R}.$$

This is a one-parameter subgroup of $\mathbb{T}^2$. It is easy to see that f is injective, differentiable, and df is injective, so f is an immersion. The image of f is thus a Lie subgroup of $\mathbb{T}^2$. We claim that it is dense in $\mathbb{T}^2$.

To see why, note that

$$f(t+k) = \left(e^{2\pi i t}, e^{2\pi i \sqrt{2}(t+k)}\right) \quad \text{for any } k \in \mathbb{Z}.$$

It suffices to show that $\left\{e^{2\pi i \sqrt{2}k}\right\}_{k\in\mathbb{Z}}$ is dense in $\mathbb{T}$, or equivalently, $\left\{\sqrt{2}k\right\}_{k\in\mathbb{Z}}$ is dense in $[0,1]$ modulo $\mathbb{Z}$. For this, we observe that the sequence $\sqrt{2}k - \left\lfloor \sqrt{2}k \right\rfloor$ is a set of distinct numbers in $[0,1]$, and so must have an accumulation point. By taking the difference of two suitable points in this sequence, we see that $\left\{\sqrt{2}k\right\}_{k\in\mathbb{Z}}$ can be arbitrarily close to 0 modulo $\mathbb{Z}$. By taking a suitable multiple, we conclude that $\left\{\sqrt{2}k\right\}_{k\in\mathbb{Z}}$ can be arbitrarily close to any fixed number in $[0,1]$ modulo $\mathbb{Z}$.

More generally, for $n \geq 2$, the one-parameter subgroup

$$t \mapsto \left(e^{2\pi i \lambda_1 t}, \ldots, e^{2\pi i \lambda_n t}\right) \in \mathbb{T}^n$$

is dense in $\mathbb{T}^n$ if $\lambda_1, \ldots, \lambda_n$ are linearly independent over $\mathbb{Q}$.

Theorem 2.2. Let $\Phi : G \to G'$ be a Lie group homomorphism, and let $\phi = (d\Phi)_e : \mathfrak{g} \to \mathfrak{g}'$. Then, the following hold:

- **(i)** Φ has constant rank k for some $k \in \mathbb{Z}$
- **(ii)** $\ker(\Phi)$ is a closed Lie subgroup of G with Lie algebra $\ker(\phi)$. In particular, $\ker(\Phi)$ has codimension k in G.
- **(iii)** There exists a neighbourhood U of e in G such that $\Phi(U)$ is an embedded submanifold of G' of dimension k

Note that if G is compact, then $\Phi(G)$ is compact and hence closed in G'; in that case the image is a closed subgroup, hence a submanifold.

Theorem 2.3. Let G be a Lie group with Lie algebra $\mathfrak{g}$. For every Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, there exists a unique connected Lie subgroup $H \subseteq G$ whose Lie algebra is $\mathfrak{h}$. In fact,

$$H = \langle \exp(\mathfrak{h}) \rangle,$$

the subgroup of G generated by $\exp(\mathfrak{h})$. Consequently, connected Lie subgroups of G are in bijection with Lie subalgebras of $\mathfrak{g}$.

Thus, fix a Lie group G with a Lie algebra $\mathfrak{g}$. Then, there is a 1-1 correspondence between connected Lie subgroups of G and Lie subalgebras of $\mathfrak{g}$.

Assume G is connected and simply connected, and let $\phi : \mathfrak{g} \to \mathfrak{g}'$ be a Lie algebra homomorphism. Consider the graph

$$\operatorname{graph}(\phi) = \{(X, \phi(X)) : X \in \mathfrak{g}\} \subseteq \mathfrak{g} \times \mathfrak{g}'.$$

Then ϕ preserves brackets if and only if $\operatorname{graph}(\phi)$ is a Lie subalgebra of $\mathfrak{g} \times \mathfrak{g}'$. So, there is a connected Lie subgroup $H \subseteq G \times G'$ with Lie algebra $\operatorname{graph}(\phi)$.

Let $f_1 : H \to G$ and $f_2 : H \to G'$ be the restrictions of the projections. On the Lie algebra level, $(df_1)_e$ is an isomorphism, so f_1 is a covering map. If G is simply connected, this covering must be an isomorphism, hence we may define

$$\Phi := f_2 \circ f_1^{-1} : G \to G',$$

which is a Lie group homomorphism whose differential at the identity is ϕ.

2.2 Transitive Actions and Isotropy Subgroups

We recall the notion of a G-space. Suppose that G is a Lie group, and M is a differentiable manifold on which G acts differentiably. We denote the collection of G-orbits by $G \setminus M$. There is a canonical projection

$$\pi : M \to G \backslash M \quad \text{where} \quad x \mapsto Gx.$$

We say that G acts on M transitively if there is only one G-orbit in M. In this case, we call M a homogeneous space for G. Recall some basic examples from MA2202 Algebra I: G acts on G by left $(x \mapsto gx)$ and right $(x \mapsto xg^{-1})$ translation and also by conjugation $(x \mapsto gxg^{-1})$. The left and right translations are transitive actions, while the conjugation is not.

Also for a subgroup $H \leq G$, H acts on G by (the restriction of) left or right translation. Clearly, H-orbits are just the left and right cosets of H. We denote the orbit space of this right action by G/H. It is clear that the left action of G on G/H is transitive, and so G/H is a homogeneous space for G. The following result is basic and it says that quotient of G by a closed Lie subgroup exhausts all differentiable transitive actions.

Theorem 2.4. The following hold:

(i) Let G be a Lie group and H be a closed subgroup. Then G/H carries a canonical differentiable manifold structure such that

$$\pi : G \to G/H$$

is a submersion, namely the rank of $d\pi$ at every point of G is equal to $\dim(G/H)$.

(ii) Let M be a (differentiable) homogeneous space of the Lie group G. Let $p \in M$ and $G_p \subseteq G$ be the isotropy group of p, then the map

$$G/G_p \to M \quad \text{where} \quad gG_p \mapsto gp$$

is a diffeomorphism.

Example 2.4. $G = O(n)$ acts on $\mathbb{R}^n$ linearly, and the action is transitive when restricted to the sphere

$$S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}.$$

The isotropy group of the south pole $(0,\ldots,0,1)$ is $O(n-1)$ (embedded in $O(n)$ in the upper left corner), so we have $O(n)/O(n-1) \cong S^{n-1}$. Similarly, $O(n)/\left(O(k) \times O(n-k)\right)$ is diffeomorphic to the manifold of all k-dimensional linear subspaces of $\mathbb{R}^n$ (called a Grassmann manifold).

Note that if H is a normal Lie subgroup of G, then the Lie algebra $\mathfrak{h}$ of H is an ideal of $\mathfrak{g}$ in the sense that

$$[X,Y] \in \mathfrak{h} \quad \text{for } X \in \mathfrak{g},\ Y \in \mathfrak{h}.$$

Next, suppose G is connected. Then there is 1-1 correspondence between connected normal Lie subgroups of G and ideals of $\mathfrak{g}$. This is because H is normal just means that H is stable under the conjugation action of G. For connected H, this means that $\mathfrak{h}$ is stable under the Adjoint action of G, and since G is connected, it is equivalent to the fact that $\mathfrak{h}$ is stable under the adjoint action of $\mathfrak{g}$.

Lastly, if H is a closed normal Lie subgroup of G, then G/H is a Lie group.

Chapter 3

Compact Lie Groups and Compact Lie Algebras

3.1 Compact Lie Groups

Definition 3.1. Let G be a compact Lie group with Lie algebra $\mathfrak{g}$.

(i) A maximal torus of G is a torus which is not properly contained in another torus in G

(ii) A maximal Abelian subalgebra of $\mathfrak{g}$ is an Abelian Lie subalgebra which is not properly contained in another Abelian Lie subalgebra

Since an Abelian Lie subalgebra is just a vector space with zeor brackets, it is contained in a maximal Abelian Lie subalgebra. To have a corresponding statement on the torus, we recall that there is a 1-1 correspondence between connected Lie subgroups of G and Lie subalgebras of $\mathfrak{g}$. This correspondence sends connected Abelian Lie subgroups of G to Abelian Lie subalgebras of $\mathfrak{g}$, and vice versa. As such, we have the following:

Proposition 3.1. The following hold:

(i) Any torus in G is contained in a maximal torus

(ii) A connected Lie subgroup of G is a maximal torus if and only if its Lie algebra is a maximal Abelian Lie subalgebra

Example 3.1. Here is an example of a maximal torus. Let $G = U(n)$, and let

$$T = \left\{ d\left(e^{i\theta_1}, \ldots, e^{i\theta_n}\right) : \theta_i \in \mathbb{R} \right\},$$

where $d\left(e^{i\theta_1}, \ldots, e^{i\theta_n}\right)$ denotes the diagonal matrix with the obvious diagonal entries. We observe that if $g \in U(n)$ and g commutes with elements of T, then g must be diagonal and so belongs to T. Thus, T is a maximal torus.

Example 3.2. We give another example of a maximal torus. Let $G = \mathrm{SO}(2n+1)$. Then, let

$$T = \left\{ \begin{pmatrix} R(\theta_1) & & & \\ & \ddots & & \\ & & R(\theta_n) & \\ & & & 1 \end{pmatrix} : \theta_i \in \mathbb{R} \right\} \quad \text{where } R(\theta) \text{ is a } 2\times 2 \text{ rotation matrix.}$$

A similar argument (replacing diagonal by block-diagonal) shows that T is a maximal torus in $SO(2n+1)$.

Definition 3.2. An inner prouct $\langle \cdot, \cdot \rangle$ on $\mathfrak{g}$ is called

(i) G-ivnariant if for all $g \in G$ and $Y, Z \in \mathfrak{g}$,

$$\langle \mathrm{Ad}(g)Y, \mathrm{Ad}(g)Z \rangle = \langle Y, Z \rangle$$

(ii) $\mathfrak{g}$-invariant if for all $X, Y, Z \in \mathfrak{g}$,

$$\langle \mathrm{ad}(X)Y, Z \rangle + \langle Y, \mathrm{ad}(X)Z \rangle = 0$$

We briefly introduce the concept of a Haar measure. Let G be a locally compact group. A measure $d\mu$ on G is called left-invariant if for $f \in L^1(G, d\mu)$,

$$\int_G f(yx)\, d\mu(x) = \int_G f(x)\, d\mu(x) \quad \text{for all } y \in G.$$

Similarly one defines right-invariant measures. A left-invariant (resp. right-invariant) measure is called a left (resp. right) Haar measure.

Theorem 3.1. Every locally compact topological group G has a left (or right) Haar measure, unique up to scaling by a positive constant.

We will not prove this theorem. When G is a Lie group, one can construct such a measure as follows: identify $\mathfrak{g}$ with left-invariant vector fields, choose a basis of partial derivatives of $\mathfrak{g}$, and let $\{dx_1, \ldots, dx_n\}$ be the dual basis of left-invariant 1-forms. Then the wedge product

$$dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n$$

is a left-invariant volume form, which defines a left-invariant measure by integration.

Definition 3.3. A locally compact topological group is called unimodular if a left Haar measure is also right-invariant.

Proposition 3.2. A compact topological group is unimodular.

Thus on a compact group G the Haar measure can be chosen both left- and right-invariant. It is customary to normalise it so that the total measure of G equals 1.

Proposition 3.3. The following hold:

(i) There exists a G-invariant inner product on $\mathfrak{g}$

(ii) Any G-invariant inner product is $\mathfrak{g}$-invariant

Proof. We first prove **(i)**. Start with any inner product $\langle \cdot, \cdot \rangle$ and define a new bilinear form

$$\langle X, Y \rangle = \int_G \langle \mathrm{Ad}(g) X, \mathrm{Ad}(g) Y \rangle \, dg,$$

where dg denotes the normalised Haar measure on G. By the invariance of Haar measure, one checks that $\langle \cdot, \cdot \rangle$ is G-invariant.

We then prove **(ii)**. Take $g = \exp(tX)$ with $X \in \mathfrak{g}$. Then for all $Y, Z \in \mathfrak{g}$ and $t \in \mathbb{R}$, we have

$$\langle \mathrm{Ad}(\exp(tX)) Y, \mathrm{Ad}(\exp(tX)) Z \rangle = \langle Y, Z \rangle .$$

Differentiate at $t = 0$, and the result follows. □

Definition 3.4. For $X \in \mathfrak{g}$, set

$$\mathfrak{g}^X = \ker(\mathrm{ad}(X)) = \{Y \in \mathfrak{g} : [X, Y] = 0\}$$

and

$$\mathfrak{g}_X = \mathrm{im}(\mathrm{ad}(X)) = \{Z \in \mathfrak{g} : Z = [X, Y], Y \in \mathfrak{g}\} .$$

By the Jacobi identity (1.7), $\mathrm{ad}(X)$ is a derivation. That is to say,

$$\mathrm{ad}(X)[Y, Z] = [\mathrm{ad}(X) Y, Z] + [Y, \mathrm{ad}(X) Z] \quad \text{where } Y, Z \in \mathfrak{g}.$$

In particular, $\mathfrak{g}^X$ is a Lie subalgebra of $\mathfrak{g}$. Next, for the adjoint action of G on $\mathfrak{g}$, the stabiliser of X is the Lie subgroup

$$G^X = \{g \in G : \mathrm{Ad}(g) X = X\} .$$

Its Lie algebra is $\mathfrak{g}^X$.

Lemma 3.1. With respect to any G-invariant inner product, there is an orthogonal direct sum decomposition as real vector spaces as follows:

$$\mathfrak{g} = \mathfrak{g}^X \oplus \mathfrak{g}_X$$

Proof. Using the G-invariance of $\langle \cdot, \cdot \rangle$, one checks that $\mathfrak{g}^X$ is orthogonal to $\mathfrak{g}_X$. By considering their dimensions, it follows that $\mathfrak{g}^X$ and $\mathfrak{g}_X$ are complementary subspaces. □

Theorem 3.2. Let $X \in \mathfrak{g}$. Then for any $Y \in \mathfrak{g}$, the $\mathrm{Ad}(G)$-orbit of Y meets $\mathfrak{g}^X$.

Definition 3.5. An element $X \in \mathfrak{g}$ is regular if for all $Y \in \mathfrak{g}$, $\dim \mathfrak{g}^X \leq \dim \mathfrak{g}^Y$.

Proposition 3.4. If X is regular, then $\mathfrak{g}^X$ is Abelian.

Proof. Let $Y \in \mathfrak{g}^X$ and consider $ad(X+tY)$ for $t \in \mathbb{R}$. Since $[X,Y]=0$, we have $ad(X+tY)$ commuting with $ad(X)$, hence it preserves the decomposition

$$\mathfrak{g} = \mathfrak{g}^X \oplus \mathfrak{g}_X.$$

As a linear map on $\mathfrak{g}_X$, the map $ad(X)$ is injective (by definition of $\mathfrak{g}^X$), hence bijective on $\mathfrak{g}_X$. Therefore for t small, $ad(X+tY)$ is also bijective on $\mathfrak{g}_X$, so

$$\mathfrak{g}_X \subset \mathfrak{g}_{X+tY}.$$

Taking orthogonal complements yields

$$\mathfrak{g}^{X+tY} \subset \mathfrak{g}^X.$$

If $Z \in \mathfrak{g}^X$ and $[Z,Y] \neq 0$, then $Z \notin \mathfrak{g}^{X+tY}$, hence

$$\dim \mathfrak{g}^{X+tY} < \dim \mathfrak{g}^X,$$

contradicting the minimality of $\dim \mathfrak{g}^X$ (regularity). Thus $[Z,Y]=0$ for all $Z \in \mathfrak{g}^X$, so $\mathfrak{g}^X$ is Abelian. □

Definition 3.6. A Cartan subalgebra is a Lie subalgebra of the form $\mathfrak{g}^X$ for some regular X.

Definition 3.7. A connected subgroup H of G is called a Cartan subgroup if its Lie algebra is a Cartan subalgebra.

By Proposition 3.4, a Cartan subgroup is connected and Abelian, hence a torus. One also checks that a Cartan subgroup can be described as the identity component of G^X for regular X, and in particular it is closed.

Theorem 3.3. If T is a maximal torus and H is a Cartan subgroup, then there exists $g \in G$ such that $gTg^{-1} = H$.

Note that in a compact Lie group, maximal tori and Cartan subgroups coincide; consequently maximal Abelian subalgebras and Cartan subalgebras coincide as well.

Proof. Choose $u \in T$ whose generated subgroup is dense in T. Then, there exists $Y \in$ (the Lie algebra of T) with $\exp(Y) = u$. As such, the $Ad(G)$-orbit of Y meets (the Lie algebra of H). Thus there exists $g \in G$ such that $Ad(g)Y \in$. Then

$$gug^{-1} = g\exp(Y)g^{-1} = \exp(Ad(g)Y) \in H,$$

so $gTg^{-1} \subset H$. Since both T and H are tori and T is maximal, this forces $gTg^{-1} = H$. □

Corollary 3.1. Any two maximal tori in G are conjugate.

Theorem 3.4. If G is a compact connected Lie group, then any $x \in G$ is conjugate to an element of a fixed maximal torus.

Proof. Since G is compact, it admits a G-invariant Riemannian metric and is complete. Because G is connected, any x can be joined to e by a geodesic; in this context the geodesic is a one-parameter subgroup through x. Hence

$$x = \exp(X) \quad \text{for some } X \in \mathfrak{g}.$$

The closure of $\{\exp(tX)\}_{t\in\mathbb{R}}$ is a compact connected abelian Lie subgroup, hence a torus. It lies in some maximal torus, and by conjugacy of maximal tori it can be conjugated into the fixed maximal torus. □

3.2 Compact Lie Algebras

Let $\mathfrak{g}$ be a (real or complex) Lie algebra. Recall that a Lie subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$ is called an ideal if $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$. It is immediate that the center

$$C(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, \mathfrak{g}] = 0\}$$

and the commutator

$$[\mathfrak{g}, \mathfrak{g}] = \{[X, Y] \mid X \in \mathfrak{g}, Y \in \mathfrak{g}\}$$

are ideals in $\mathfrak{g}$.

Definition 3.8 (compact Lie algebra). A real Lie algebra is called a compact Lie algebra if it is the Lie algebra of a compact Lie group.

For the rest of this section, fix a compact Lie algebra $\mathfrak{g}$. Recall that there exists a $\mathfrak{g}$-invariant inner product on $\mathfrak{g}$. Fix such an inner product $\langle \cdot, \cdot \rangle$.

Lemma 3.2. With respect to $\langle \cdot, \cdot \rangle$, we have the orthogonal direct sum of ideals

$$\mathfrak{g} = C(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}].$$

Furthermore, if we denote $K = [\mathfrak{g}, \mathfrak{g}]$, then

$$C(K) = 0 \quad \text{and} \quad [K, K] = K.$$

Proof. Let $X, Y, Z \in \mathfrak{g}$. Then,

$$\langle [X, Y], Z \rangle = -\langle Y, [X, Z] \rangle.$$

Clearly,

$$\begin{aligned} Z \in C(\mathfrak{g}) \quad & \text{if and only if} \quad [Z,X] \text{ for all } X \in \mathfrak{g} \\ & \text{if and only if} \quad \langle Y,[Z,X]\rangle = 0 \text{ for all } X,Y \in \mathfrak{g} \end{aligned}$$

This is equivalent to $\langle [X,Y],Z\rangle = 0$ for all $X,Y \in \mathfrak{g}$. That is to say, $Z \perp [\mathfrak{g},\mathfrak{g}]$. Hence, the orthogonal complement of $C(\mathfrak{g})$ is $[\mathfrak{g},\mathfrak{g}]$. Since $\mathfrak{g} = C(\mathfrak{g}) \oplus K$, we have $K = [\mathfrak{g},\mathfrak{g}] = [K,K]$. Also, $C(K) \subseteq C(\mathfrak{g})$, so $C(K) = \{0\}$. □

Definition 3.9 (simple Lie algebra). A Lie algebra $\mathfrak{g}$ is called simple if $\mathfrak{g}$ is not Abelian and $\mathfrak{g}$ has no non-trivial ideals.

If $\mathfrak{g}$ is Abelian, then every subspace of $\mathfrak{g}$ is an ideal.

Example 3.3. We show that $\mathfrak{g} = \mathfrak{su}(2)$ is a simple real Lie algebra. The Lie algebra $\mathfrak{su}(2)$ consists of 2×2 skew-Hermitian matrices of trace zero and has the basis $\{J_1, J_2, J_3\}$, where

$$J_1 = \frac{1}{2}\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} \quad \text{and} \quad J_2 = \frac{1}{2}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \quad \text{and} \quad J_3 = \frac{1}{2}\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}.$$

We may easily check that

$$[J_1,J_2] = J_3 \quad [J_2,J_3] = J_1 \quad [J_3,J_1] = J_2.$$

Suppose I is an ideal in $\mathfrak{su}(2)$ and $aJ_1 + bJ_2 + cJ_3$ is a non-zero element of I. Assume $a \neq 0$. Applying $\operatorname{ad}(J_3)$ gives $aJ_2 - bJ_1 \in I$. Applying $\operatorname{ad}(J_1)$ yields $aJ_3 \in I$, hence $J_3 \in I$. Then $J_1, J_2 \in I$, so $I = \mathfrak{su}(2)$. Thus $\mathfrak{su}(2)$ is simple.

Definition 3.10 (semisimple Lie algebra). Let $\mathfrak{g}$ be a Lie algebra over a field F. We say $\mathfrak{g}$ is semisimple if it is a direct sum of simple Lie algebras.

We then discuss the Killing form, which is a symmetric bilinear form that plays a basic role in the theories of Lie groups and Lie algebras. Cartan's criteria (criterion of solvability and criterion of semisimplicity in Theorem 3.6) show that Killing form has a close relationship to the semisimplicity of the Lie algebras.

Definition 3.11 (Killing form). The Killing form of $\mathfrak{g}$ is

$$K(X,Y) = \operatorname{tr}(\operatorname{ad}(X)\operatorname{ad}(Y)), \quad \text{where } X,Y \in \mathfrak{g}.$$

It is bilinear and symmetric.

Proposition 3.5. The Killing form is $\mathfrak{g}$-invariant.

Proof. For $X,Y,Z \in \mathfrak{g}$, we have

$$K([X,Y],Z) = \operatorname{tr}(\operatorname{ad}[X,Y]\operatorname{ad}(Z)) = \operatorname{tr}((\operatorname{ad}(X)\operatorname{ad}(Y) - \operatorname{ad}(Y)\operatorname{ad}(X))\operatorname{ad}(Z)).$$

By the cyclic property of trace, this equals to $K(Y,[X,Z])$. □

Lemma 3.3. Let $\mathfrak{g}$ be a compact Lie algebra. Then the Killing form K is negative semidefinite, and

$$K(X,X) = 0 \quad \text{if and only if} \quad X \in C(\mathfrak{g}).$$

Proof. Since $\langle \cdot, \cdot \rangle$ is $\mathfrak{g}$-invariant, for each $X \in \mathfrak{g}$, the operator $\operatorname{ad}(X)$ is skew-adjoint and hence diagonalisable over $\mathbb{C}$ with purely imaginary eigenvalues $i\lambda_j$. Hence, $(\operatorname{ad}(X))$ has eigenvalues $-\lambda_j^2$, and

$$K(X,X) = \operatorname{tr}\left(\operatorname{ad}(X)^2\right) = \sum_j \left(-\lambda_j^2\right) \leq 0.$$

Moreover, $K(X,X) = 0$ iff all $\lambda_j = 0$, i.e. $\operatorname{ad}(X) = 0$, i.e. $X \in C(\mathfrak{g})$. □

Note that for a compact Lie algebra $\mathfrak{g}$, its commutator ideal $[\mathfrak{g}, \mathfrak{g}]$ is semisimple, and

$$\mathfrak{g} \text{ is semisimple} \quad \text{if and only if} \quad C(\mathfrak{g}) = 0.$$

Corollary 3.2. If $\mathfrak{g}$ is a semisimple compact Lie algebra, then its Killing form is negative definite.

Definition 3.12. For a Lie algebra $\mathfrak{g}$, define the following:

(i) $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}]$, $\mathfrak{g}^{(2)} = \left[\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}\right]$, and $\mathfrak{g}^{(i)} = \left[\mathfrak{g}^{(i-1)}, \mathfrak{g}^{(i-1)}\right]$

(ii) $\mathfrak{g}_0 = \mathfrak{g}$, $\mathfrak{g}_1 = [\mathfrak{g}, \mathfrak{g}]$, $\mathfrak{g}_2 = [\mathfrak{g}, \mathfrak{g}_1]$, and $\mathfrak{g}_i = [\mathfrak{g}, \mathfrak{g}_{i-1}]$

The sequence $\mathfrak{g}^{(i)}$ is the derived series of $\mathfrak{g}$, and $\mathfrak{g}_i$ is the lower central series. We say $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(k)} = 0$ for some k, and nilpotent if $\mathfrak{g}_k = 0$ for some k. By induction, $\mathfrak{g}^{(i)} \subseteq \mathfrak{g}_i$, so nilpotent implies solvable.

Example 3.4. Let $\mathfrak{b} \subseteq \mathfrak{gl}(n, \mathbb{R})$ be the Lie algebra of upper triangular matrices and $\mathfrak{n} \subseteq \mathfrak{b}$ the strictly upper triangular matrices. Then $\mathfrak{b}$ is solvable but not nilpotent, and $\mathfrak{n}$ is both solvable and nilpotent.

Example 3.5. Let V be a real vector space with a symplectic form $\omega : V \times V \to \mathbb{R}$. Let $G = V \times \mathbb{R}$ with group operation

$$(v_1, \alpha_1) \cdot (v_2, \alpha_2) = \left(v_1 + v_2, \alpha_1 + \alpha_2 + \frac{1}{2}\omega(v_1, v_2)\right).$$

Then $\mathfrak{g} = V \times \mathbb{R}$ and the Lie bracket is given by the corresponding commutator.

Definition 3.13. Given a Lie algebra $\mathfrak{g}$, there exists a maximal solvable ideal of $\mathfrak{g}$ called the radical of $\mathfrak{g}$, denoted by $\operatorname{rad}(\mathfrak{g})$.

Lemma 3.4. Let $\mathfrak{g}$ be a compact Lie algebra and $\mathfrak{h}$ an Abelian ideal of $\mathfrak{g}$. Then $\mathfrak{h} \subseteq C(\mathfrak{g})$.

Proof. Let $X \in \mathfrak{h}$ and $Y \in \mathfrak{g}$. Then $\operatorname{ad}(X)$ maps $\mathfrak{g}$ into $\mathfrak{h}$, and since $\mathfrak{h}$ is Abelian we have $\operatorname{ad}(X)^2 = 0$. Hence

$$K(X,X) = \operatorname{tr}\left(\operatorname{ad}(X)^2\right) = 0,$$

so $X \in C(\mathfrak{g})$. □

Theorem 3.5. A compact Lie algebra $\mathfrak{g}$ splits (uniquely) into the direct sum of its center and simple ideals:

$$\mathfrak{g} = C(\mathfrak{g}) \oplus K_1 \oplus K_2 \oplus \cdots \oplus K_\ell,$$

where the K_i are the simple ideals of $K = [\mathfrak{g}, \mathfrak{g}]$.

Proof. We proceed by induction on $\dim \mathfrak{g}$. If $\dim \mathfrak{g} = 1$, then $\mathfrak{g} = C(\mathfrak{g})$. Assume $\dim \mathfrak{g} > 1$.

If $C(\mathfrak{g}) \neq 0$, then $\dim K < \dim \mathfrak{g}$ and we apply the induction hypothesis to K. If $C(\mathfrak{g}) = 0$, let $\mathfrak{g}_1$ be a minimal non-zero ideal of $\mathfrak{g}$. It is not Abelian (otherwise $\mathfrak{g}_1 \subseteq C(\mathfrak{g}) = 0$), hence it is simple. We claim $\mathfrak{g}_1^\perp$ is also an ideal of $\mathfrak{g}$. Indeed, let $X \in \mathfrak{g}_1^\perp$. For any $Y \in \mathfrak{g}$ and $Z \in \mathfrak{g}_1$,

$$\langle [X,Y], Z \rangle = -\langle X, [Y,Z] \rangle = 0,$$

so $[X,Y] \in \mathfrak{g}_1^\perp$. Thus

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_1^\perp.$$

Apply the induction hypothesis to $\mathfrak{g}_1^\perp$, noting that any ideal of $\mathfrak{g}_1^\perp$ is an ideal of $\mathfrak{g}$. □

Corollary 3.3. Let $\mathfrak{g}$ be a compact Lie algebra. Then

$$\mathfrak{g} \text{ is semisimple if and only if } C(\mathfrak{g}) = 0 \text{ if and only if } \operatorname{rad}(\mathfrak{g}) = 0.$$

Thus $\mathfrak{g}$ is semisimple if and only if its radical is zero (this is the standard definition in many references).

For a general Lie algebra (not necessarily compact), we state the following important result known as Cartan's criteria (Theorem 3.6):

Theorem 3.6 (Cartan's criteria). Let $\mathfrak{g}$ be a Lie algebra over a field of characteristic 0. Then

(i) $\mathfrak{g}$ is solvable if and only if its Killing form is 0

(ii) $\mathfrak{g}$ is semisimple if and only if its Killing form is non-degenerate.

Given a Lie algebra homomorphism $\phi : \mathfrak{g} \to \mathfrak{gl}(V)$, we may define a bilinear symmetric form K_ϕ by

$$K_\phi(X,Y) = \operatorname{tr}(\phi(X)\phi(Y)) \quad \text{where } X, Y \in \mathfrak{g}.$$

Then K_ϕ is $\mathfrak{g}$-invariant. The Killing form is the special case $\phi = \operatorname{ad}$.

Chapter 4

Representation Theory of Compact Groups

4.1 Characters as Class Functions

Let G be a compact topological group. We fix a Haar measure on G, denoted by dx, such that the total measure of G is 1.

Definition 4.1. A finite-dimensional representation of G is a continuous homomorphism

$$\pi : G \to \mathrm{GL}(V),$$

where V is a finite-dimensional complex vector space.

A subspace $V_0 \subseteq V$ is called invariant if

$$\pi(x) V_0 \subseteq V_0 \quad \text{for all } x \in G.$$

The spaces $\{0\}$ and V are called trivial invariant subspaces. Lastly, π is said to be irreducible if it has no non-trivial invariant subspaces.

When V is endowed with a positive definite Hermitian form $\langle \cdot, \cdot \rangle$ (or an inner product), a continuous homomorphism $\pi : G \to U(V)$ is called a unitary representation. Here, $U(V)$ denotes the group of unitary operations with respect to the form $\langle \cdot, \cdot \rangle$.

Remark: In a similar way we may also talk about infinite-dimensional representations, usually in Hilbert spaces $\mathcal{H}$. To avoid technicalities, we limit ourselves to unitary representations $\pi : G \to U(\mathcal{H})$, and we use the strong topology on $U(\mathcal{H})$ to specify continuity of π. Also we say π is irreducible if there are no non-trivial closed invariant subspaces in $\mathcal{H}$.

Note that finite-dimensional representations are automatically smooth. This follows from

Cartan's closed subgroup criterion (Theorem 3.6). If $f : G \to H$ is a continuous homomorphism between Lie groups then its graph

$$\Gamma_f = \{(x, f(x)) : x \in G\} \subseteq G \times H$$

is a closed subgroup and hence an embedded submanifold. The projection maps

$$\pi_1 : G \times H \to G \quad \text{and} \quad \pi_2 : G \times H \to H$$

are both smooth and so $\pi_1 |_{\Gamma_f}$ is smooth bijection. Its derivative is an isomorphism and so by the constant rank theorem, it is a local diffeomorphism everywhere and hence a diffeomorphism. The map f can then be recovered as the composition $\pi_2 \circ \pi_1^{-1}$ which is smooth.

Definition 4.2 (representation of Lie algebra). A finite-dimensional representation of $\mathfrak{g}$ is a Lie algebra homomorphism

$$\pi : \mathfrak{g} \to \mathfrak{gl}(V)$$

i.e. $\pi([x,y]) = [\pi(x), \pi(y)]$ for all $x, y \in \mathfrak{g}$.

For a connected and simply connected Lie group G, there is a bijection between the sets of finite-dimensional representations of G and finite-dimensional representations of $\mathfrak{g}$.

Lemma 4.1. Suppose that (π, V) is a finite-dimensional representation. Then there exists a positive definite Hermitian form such that π is unitary with respect to this form.

Proof. Take any positive definite Hermitian form $(\cdot,\cdot)$ on V. Then, define a new form $\langle \cdot,\cdot \rangle$ by averaging as follows:

$$\langle u, v \rangle = \int_G (\pi(x)u, \pi(x)v)\, dx.$$

Then, clearly π is unitary with respect to $\langle \cdot,\cdot \rangle$. □

Proposition 4.1. Let π be a finite dimensional representation of G, then V is a direct sum of irreducible (sub)representations.

Proof. We may assume that π is unitary. We claim that if V_0 is an invariant subspace, then the orthogonal complement $V_0^\perp$ of V_0 is also invariant. To see this, let $u \in V_0^\perp$, then for any $v \in V_0$, then

$$\langle \pi(g)u, v \rangle = \langle u, \pi(g^{-1})v \rangle = 0.$$

We continue the proof. If π is irreducible, then there is nothing to prove. If not, let V_0 be a non-trivial invariant subspace, then we have

$$V = V_0 \oplus V_0^\perp,$$

as representations of G. We then continue (for a finite number of times) until we get a direct sum of irreducible representations. □

Definition 4.3 (intertwining representation). Given two representations $\pi : G \to \mathrm{GL}(V_i)$ for $i = 1,2$, an operator $T \in \mathrm{Hom}(V_1, V_2)$ is called intertwining if

$$T\pi_1(g) = \pi_2(g)T \quad \text{for all } g \in G.$$

The space of intertwining operators is denoted by $\mathrm{Hom}_G(V_1, V_2)$.

Definition 4.4 (equivalent representation). Two representations $\pi : G \to \mathrm{GL}(V_i)$ for $i = 1,2$ are called equivalent if there exists an invertible intertwining map T, namely

$$T\pi_1(g)T^{-1} = \pi_2(g) \quad \text{for all } g \in G.$$

We write $\pi_1 \cong \pi_2$. Representations which are not equivalent are said to be inequivalent.

Lemma 4.2 (Schur's lemma). The following hold:

(i) Given two irreducible representations $\pi_i : G \to \mathrm{GL}(V_i)$ $(i = 1,2)$ and $T \in \mathrm{Hom}_G(V_1, V_2)$, then T is either 0 or invertible. Consequently if π_1 and π_2 are inequivalent, then

$$\mathrm{Hom}_G(V_1, V_2) = 0.$$

(ii) If $T \in \mathrm{Hom}_G(V_1, V_1)$, then $T = \lambda I$ for some $\lambda \in \mathbb{C}$. Consequently if π_1 and π_2 are equivalent, then

$$\dim \mathrm{Hom}_G(V_1, V_2) = 1.$$

Corollary 4.1. Write

$$V \cong \sum_i m_i V_i$$

as direct sums of inequivalent irreducible representations V_i, where $m_i \in \mathbb{Z}^+$ are non-negative integers. Then for each i, $m_i = \dim \mathrm{Hom}_G(V_i, V)$.

In Corollary 4.1, the integer m_i is called the multiplicity of V_i in V.

Proof. We have

$$\mathrm{Hom}_G(V_i, V) = \mathrm{Hom}_G\left(V_i, \sum_j m_j V_j\right) \cong \sum_j m_j \mathrm{Hom}_G(V_i, V_j),$$

and by Schur's lemma (Lemma 4.2), its dimension is equal to m_i. □

Proposition 4.2. A finite-dimensional representation (π, V) of a compact topological group G is irreducible if and only if $\mathrm{Hom}_G(V,V) = \mathbb{C}I$.

Proof. Write $V \cong \sum_i m_i V_i$, then

$$\mathrm{Hom}_G(V,V) = \sum_{i,j} m_i m_j \mathrm{Hom}_G(V_i, V_j),$$

and so

$$\dim \mathrm{Hom}_G(V,V) = \sum_i m_i^2.$$

This dimension is 1 if and only if one of m_i is 1 and the rest is 0, i.e. V is irreducible. □

We look at the special case where G is Abelian.

Proposition 4.3. If G is abelian, and (π, V) is an irreducible finite-dimensional representation of G, then V is one dimensional.

A 1-dimensional representation is called a character: it is just a continuous homomorphism

$$G \to \mathbb{C}^\times.$$

Example 4.1. Let $G = \mathbb{T}^n$, the k-dimensional torus. For each $(\lambda_1, \ldots, \lambda_n) \in \mathbb{Z}^n$, we may define a unitary character of $\mathbb{T}^n$ by the map

$$\left(e^{it_1}, \ldots, e^{it_n}\right) \mapsto e^{i\lambda_1 t_1} \cdots e^{i\lambda_n t_n}.$$

In fact, all (unitary) characters of $\mathbb{T}^n$ are of this form.

Definition 4.5 (dual representation). Let (π, V) be a representation. The dual representation or he contragradient representation (π^*, V^*) is defined as follows:V^* is the dual space of V, and

$$(\pi^*(g)v^*)(v) = v^*(\pi(g)^{-1}v) \quad \text{where } g \in G, v^* \in V^*, v \in V.$$

Definition 4.6 (tensor product representation). Given two representations (π_i, V_i), $i = 1, 2$, the tensor product representation $(\pi_1 \otimes \pi_2, V_1 \otimes V_2)$ is defined by

$$(\pi_1 \otimes \pi_2)(g)(v_1 \otimes v_2) = \pi_1(g)v_1 \otimes \pi_2(g)v_2 \quad \text{where } g \in G,\ v_1 \in V_1,\ v_2 \in V_2.$$

Note that for a linear operator A on a vector space V, its trace may be computed using matrix representation of A in any basis:

$$\mathrm{Tr}(A) = \sum_i v_i^*(Av_i),$$

where $\{v_i\}$ is a basis of V and $\{v_i^*\}$ its dual basis. Note that $v_i^*(Av_i)$ is the coefficient of Av_i in front of v_i.

We now fix a compact topological group G.

Definition 4.7 (character of representation). Let (π, V) be a representation. The character of π, denoted by χ_π, is the continuous function on G given by

$$\chi_\pi(g) = \mathrm{Tr}(\pi(g)) \quad \text{where } g \in G.$$

Clearly equivalent representations have the same characters.

Proposition 4.4 (elementary properties of characters). The following hold:

(i) $\chi_\pi(e) = \dim V$

(ii) $\chi_\pi(g) = \chi_\pi(hgh^{-1})$, for $h, g \in G$. We say that χ_π is a class function.

(iii) $\chi_{\pi^*}(g) = \chi_\pi(g^{-1}) = \overline{\chi_\pi(g)}$, for $g \in G$

(iv) $\chi_{\pi_1 \otimes \pi_2}(g) = \chi_{\pi_1}(g)\chi_{\pi_2}(g)$, for $g \in G$

The follwowing lemma tells us how to produce invariants. The process is called averaging.

Lemma 4.3 (production of invariants). For a finite-dimensional representation (π, V), define the operator

$$P = \int \pi(g)\, dg.$$

Then, $\mathrm{im}(P) = V^G$, where V^G denotes the space of G-invariants.

Proof. We have

$$P^2 v = P(Pv) = P\left(\int_G \pi(x) v\, dx\right) = \int_G P(\pi(x) v)\, dx$$

which is equal to

$$\int_G \int_G \pi(y)\pi(x) v\, dy dx = \int_G \int_G \pi(yx) v\, dy dx = \int_G \int_G \pi(z) v\, dz = Pv$$

so $Pv \in V^G$. Conversely, if $w \in V^G$, then

$$Pw = \int_G \pi(x) w\, dx = \int_G w\, dx = w.$$

Thus, we conclude that $\mathrm{im}(P) = V^G$. □

One recalls from either MA2001 Linear Algebra 1 or MA2101 Linear Algebra 2 that a linear operator P on V is a projection operator onto W if $P^2 = P$ and $W = \mathrm{im}(P)$. We have $\mathrm{Tr}(P) = \dim W$. To see this, fix a basis $\{w_1, \ldots, w_k\}$ of W and extend it to a basis $\{w_1, \ldots, w_k, w_{k+1}, \ldots, w_n\}$ of V. Then $Pw_i = w_i$ for $i \leq k$, and Pw_i is in the span of $\{w_1, \ldots, w_k\}$ for $i > k$, so $\mathrm{Tr}(P) = k$.

Corollary 4.2.

$$\int_G \chi_\pi(x)\, dx = \dim V^G.$$

Proof. We note that

$$\int_G \chi_\pi(x)\,dx = \int_G Tr(\pi(x))\,dx = \mathrm{Tr}(P)$$

where P is the averaging operator. Since P is a projection operator onto V^G, its trace is just $\dim V^G$. □

Theorem 4.1. Given two representations (π_i, V_i) for $i = 1,2$, we have

$$\int_G \overline{\chi_{\pi_1}(x)}\,\chi_{\pi_2}(x)\,dx = \dim Hom_G(V_1, V_2).$$

Proof. We have

$$\int_G \overline{\chi_{\pi_1}(x)}\,\chi_{\pi_2}(x)\,dx = \int_G \chi_{\pi_1^*}(x)\chi_{\pi_2}(x)\,dx = \int_G \chi_{\pi_1^*\otimes\pi_2}(x)\,dx = \dim(V_1^* \otimes V_2)^G$$

which is equal to $\dim Hom_G(V_1, V_2)$. □

Corollary 4.3 (Schur orthogonality relations). If (π_i, V_i) are irreducible for $i = 1,2$, then

$$\int_G \overline{\chi_{\pi_1}(x)}\,\chi_{\pi_2}(x)\,dx = \begin{cases} 0, & \pi_1 \not\cong \pi_2; \\ 1, & \pi_1 \cong \pi_2. \end{cases}$$

Note that we have the standard Hermitian inner product on $L^2(G)$, which is defined to be

$$\langle f_1, f_2 \rangle = \int_G f_1(x)\overline{f_2(x)} \quad \text{where } f_1, f_2 \in L^2(G).$$

Then Schur's orthogonality relations (Corollary 4.3) says that irreducible characters form an orthonormal set, and in fact as we shall see later, they form an orthonormal basis in the space of L^2 class functions on G.

Corollary 4.4. A finite-dimensional representation (π, V) of a compact topological group G is determined (up to equivalence) by its character. More precisely, if we write $V \cong \sum_i m_i V_i$ as direct sums of inequivalent irreducible representations V_i, then

$$m_i = \langle \chi_\pi, \chi_i \rangle,$$

where χ_i is the character of the irreducible representation V_i.

Theorem 4.2. Let $(\pi, \mathcal{H})$ be an irreducible unitary representation. Then any bounded linear operator commuting with $\pi(g)$, for $g \in G$, is scalar.

Proof. Suppose that L is a non-scalar bounded linear operator, and it commutes with all $\pi(g)$, $g \in G$. Since π is unitary, the adjoint of L, denoted by L^*, also commutes with all $\pi(g)$, where $g \in G$.

We write

$$L = A + iB \quad \text{where} \quad A = \frac{L + L^*}{2} \text{ and } B = \frac{L - L^*}{2i}.$$

Here, A and B are self-adjoint bounded linear operators, and one of them must be non-scalar, say A. Clearly A commutes with all $\pi(g)$, for $g \in G$. We have (by the spectral theorem) that

$$A = \int_{\mathbb{R}} \lambda \, dP_\lambda,$$

where P_λ is an orthogonal projection onto a closed subspace E_λ of $\mathcal{H}$ and $\lambda \mapsto dP_\lambda$ is a projection-valued measure. Furthermore each P_λ is a function of A.

This implies that there exists a non-scalar projection P_λ, and since P_λ is a function of A, it commutes with all $\pi(g)$, $g \in G$. One easily checks the corresponding closed subspace E_λ is then a non-trivial invariant subspace, thus contradicting the irreducibility of π. □

Definition 4.8 (matrix coefficient of representation). Let $(\pi, \mathcal{H})$ be a unitary representation of G. A matrix coefficient of π is a function of the form

$$\phi_{u,v}(x) = \langle \pi(x)u, v \rangle, \qquad x \in G,$$

where $u, v \in \mathcal{H}$.

Theorem 4.3 (Schur's orthogonality relations). Let $(\pi_i, \mathcal{H}_i)$ be irreducible unitary representations of G, where $i = 1, 2$.

(i) If $\pi_1 \not\cong \pi_2$, then

$$\int_G \langle \pi_1(x)u_1, v_1 \rangle \overline{\langle \pi_2(x)u_2, v_2 \rangle} \, dx = 0$$

for $u_1, v_1 \in \mathcal{H}_1$ and $u_2, v_2 \in \mathcal{H}_2$.

(ii) If $\pi_1 = \pi_2 = \pi$, then there exists a positive real number c (depending on π) such that

$$\int_G \langle \pi(x)u_1, v_1 \rangle \overline{\langle \pi(x)u_2, v_2 \rangle} \, dx = c \, \langle u_1, u_2 \rangle \overline{\langle v_1, v_2 \rangle},$$

for $u_1, v_1, u_2, v_2 \in \mathcal{H}$.

4.2 The Peter-Weyl Theorem

Let G be a compact group. We shall consider unitary representations of G in a Hilbert space $\mathcal{H}$. We start with the following version of Schur's lemma. Note that $\mathcal{H}$ may be infinite-dimensional, so the version of Schur's lemma in Lemma 4.2 cannot be applied directly.

Theorem 4.4. Any irreducible unitary representation $(\pi, \mathcal{H})$ of a compact group G is finite-dimensional.

Proof. We fix a unit vector $v \in \mathcal{H}$. Let $e_1, e_2, \ldots, e_n$ be an arbitrary orthonormal collection of vectors in $\mathcal{H}$. Then we have

$$\sum_{i=1}^{n} \left|\langle \pi(x)v, e_i\rangle\right|^2 \leq \|\pi(x)v\|^2 = \|v\|^2 = 1 \quad \text{where } x \in G.$$

Integrating this inequality over the group G, and using Schur's orthogonality relations, we see that $cn \leq 1$, for some $c > 0$.[1] Thus $n \leq \frac{1}{c}$. This implies that $\mathcal{H}$ is finite-dimensional. □

Definition 4.9. The left regular representation of G is the representation of G on $L^2(G)$ given by

$$(L(g)f)(x) = f(g^{-1}x) \quad \text{where } f \in L^2(G),\ g, x \in G.$$

The right regular representation is defined similarly. They are obviously unitary representations.

Definition 4.10. Let $\widehat{G}$ denote the set of equivalent classes of finite-dimensional irreducible unitary representations of G, called the unitary dual of G.

For $(\pi, V) \in \widehat{G}$, let $\{v_i\}_{1 \leq i \leq \dim \pi}$ be an orthonormal basis of V, and $\phi^{\pi}_{i,j}$ be the matrix coefficient given by

$$\phi^{\pi}_{i,j}(x) = \langle \pi(x)v_i, v_j\rangle \quad \text{where } x \in G.$$

Theorem 4.5 (Peter-Weyl theorem). The following hold:

(i) The linear span of all matrix coefficients of $\pi \in \widehat{G}$ is dense in $L^2(G)$.

(ii)

$$\left\{(\dim(\pi))^{\frac{1}{2}} \phi^{\pi}_{i,j}\right\}_{\pi \in \widehat{G},\, 1 \leq i,j \leq \dim \pi}$$

is an orthonormal basis of $L^2(G)$

(iii) We have the following decomposition of the left regular representation:

$$L^2(G) \cong \sum_{\pi \in \widehat{G}} \dim(\pi)\pi$$

Note that if G is a compact matrix group, then **(i)** is an immediate consequence of Stone-Weierstrass theorem: in fact, the linear span of all matrix coefficients forms an algebra of continuous functions closed under conjugation, containing the constants and

[1] The constant c in Schur's orthogonality relations is actually $\frac{1}{n}$, where $n = \dim \mathcal{H}$.

separating points (by using tensor products, complex conjugate, the trivial and the given matrix representation, respectively). This algebra is thus uniformly dense in the space of all continuous complex-valued functions on G.

Lemma 4.4. Let $(\pi, V) \in \widehat{G}$. We have

$$\int_G \pi(xgx^{-1})\,dx = \frac{\chi_\pi(g)}{\dim \pi} I.$$

Lemma 4.5. Consequently

$$\int_G \langle \pi(xgx^{-1})v, w\rangle\,dx = \frac{\chi_\pi(g)}{\dim \pi} \langle v, w\rangle,$$

for $v, w \in V$.

Theorem 4.6. The linear span of the characters $\{\chi_\pi \mid \pi \in \widehat{G}\}$ is dense in $L^2(G)_{class}$, the space of conjugation invariant functions in $L^2(G)$. Consequently, the characters $\{\chi_\pi \mid \pi \in \widehat{G}\}$ form an orthonormal basis for $L^2(G)_{class}$.

Corollary 4.5. Let G be a finite group. Write $\widehat{G} = \{\pi_i\}_{1 \le i \le r}$, where r is the number of equivalent classes of (finite dimensional) irreducible representations of G. Then the following hold:

(i)

$$\sum_{i=1}^{r} d_i^2 = |G|,$$

where $d_i = \dim \pi_i$. Then we have

(ii) Any class function on G is a linear combination of χ_{π_i}, for $1 \le i \le r$.

(iii) The number r is equal to the number of conjugate classes of G.

4.3 SU(2)

Chapter 5

Roots, Root Space Decomposition and Highest Weight Theory

5.1 Roots and Root Space Decomposition

5.2 Root System

5.3 Weyl Group and Basic Properties of Roots

5.4 Simple roots, Weyl Chambers, Highest Root

5.5 Cartan Matrices

5.6 Dynkin Diagrams and Classification of Root Systems

5.7 Weight Lattice, Fundamental Weights, Dominant Weights

5.8 Theorem of Highest Weight

5.9 Weyl's Formulae

5.10 Weyl's Integration Formula

5.11 Weyl's Character Formula

Bibliography

[1] Hall, B. C. (2015). *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction* (2nd ed.). Springer. ISBN: 9783319134666.